

Equal Coverings of Finite Groups: Theory, Lean Formalization, and Computational Evidence

Bishesh Bohora, Ashwot Acharya, Pranjali Man Shakya, Bhoomika Thapa

Department of Mathematics

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Abstract

An **equal covering** of a finite group G is a collection of proper subgroups of equal order whose union is G . This report documents a research extension of the work of Velasquez-Berroteran [6], which developed the theory of equal coverings and computationally classified groups up to order 60. Our work proceeds along three strands. First, we present proofs of several foundational results on group coverings that the paper invokes but does not prove in full situating these within the equal covering framework. Second, we formalize key theorems on equal coverings in Lean 4 using Mathlib, identifying proof-theoretic subtleties along the way. Third, we extend the GAP-based computational classification of groups with equal coverings from order 60 to order 120, producing a complete table of results with supporting theoretical justifications. Together, these contributions provide a more complete theoretical foundation for the subject, a formal verification layer, and new computational evidence bearing on the open conjecture that no finite simple group admits an equal covering. Furthermore, We attempted a possible direction on proving the conjecture for some Alternating Groups.

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1 Introduction

The study of representing groups as unions of proper subgroups is a classical area of group theory, originating with Scorza [5] in 1926 and later revitalized by work of Neumann, Cohn, and many others. An *equal covering* imposes the additional constraint that all subgroups in the covering have the same order, making the problem significantly more delicate.

This report documents a research extension of the thesis by Velasquez-Berroteran [6], which developed the foundational theory of equal coverings and classified groups up to order 60. We pursue three directions:

- (i) **Theoretical:** rigorous proofs theorems on group covering, including the Haber–Rosenfeld theorem [4], Cohn’s results [3], study of Alternating Groups.
- (ii) **Formal:** Lean 4 formalizations of selected results.
- (iii) **Computational:** GAP-based classification of groups of orders 61–120.

A central open problem motivating our work is the following conjecture from [6]:

Conjecture 1.1 (Velasquez-Berroteran, 2022). *If G is a finite simple group, then G has no equal covering.*

Structure of the report. Section 2 collects definitions and background. Section 3 presents the theoretical results. Section 4 describes the Lean formalization. Section 5 presents the GAP algorithm and computational results. Section 6 analyzes Theorem 1.1. Section 7 summarizes and lists future directions.

2 Preliminaries

Throughout, all groups are *finite* unless otherwise specified.

Definition 2.1 (Covering). Let G be a group. A *covering* of G is a collection $\Pi = \{H_i\}$ of proper subgroups of G such that $G = \bigcup_i H_i$. The *covering number* $\sigma(G)$ is the minimum cardinality of any covering of G ; we set $\sigma(G) = \infty$ if G is cyclic.

Definition 2.2 (Equal covering). A covering $\Pi = \{H_i\}$ of G is an *equal covering* if $|H_i| = |H_j|$ for all i, j . The *equal covering number* $\varepsilon(G)$ is the minimum cardinality of any equal covering of G ; we write $\varepsilon(G) = \infty$ if G has no equal covering.

Definition 2.3 (Irredundant covering [6, Definition 1]). A covering Π is *irredundant* (or *minimal*) if for each $H_i \in \Pi$ there exists $x_i \in H_i$ with $x_i \notin \bigcup_{j \neq i} H_j$.

Definition 2.4 (Exponent [6, Definition 4]). The *exponent* of G , written $\exp(G)$, is the smallest positive integer n such that $g^n = 1$ for all $g \in G$. For finite G , $\exp(G) = \text{lcm}\{|g| : g \in G\}$.

Definition 2.5 (Semidirect product [6, Definition 5]). G is the *semidirect product* of subgroups H and K , written $G = H \rtimes K$, if $H \trianglelefteq G$, $HK = G$, and $H \cap K = \{1\}$.

Definition 2.6 (Partition). A covering Π of G is a *partition* if any two distinct members intersect trivially. The *partition number* $\rho(G)$ is the minimum cardinality of any partition.

Remark 2.7. Every equal partition is an equal covering, so $\varepsilon(G) \leq \rho(G)$ whenever both are finite. However, the classes of groups with equal coverings and equal partitions differ substantially.

3 Theoretical Results

3.1 Fundamental Results in Group Covering

Before studying equal coverings, we worked through the following foundational results on general group coverings. The following proofs are included to document our study of the theory and also because cannot be found directly in [6].

Theorem 3.1 ([6, Theorem 1]). *A group G has a covering if and only if G is non-cyclic.*

Proof. Suppose G has a covering Π and G is cyclic, Let g be the generator of G i.e $\langle g \rangle = G$, Since Π is a covering of G , $g \in H$, $H \in \Pi$. Since g generates G , $g \in H \implies G = H$. Hence H is not proper and no such covering exist.

Conversly, Suppose G is non cyclic, then define $\Pi = \{\langle a \rangle \mid a \in G\}$ Since G is non cyclic, each $\langle a \rangle$ is a proper subgroup.

□

Theorem 3.2 ([6, Theorem 2]). *For any group G , $\sigma(G) \neq 2$.*

It is a fundamental result in Group Theory That a group cannot be expressed as union of two proper subgroups, We again prove this result.

Proof. Suppose $H, K < G$ and $H \cup K = G$. Let $h \in H$ and $k \in K$, $\implies hk \in G \implies hk \in H \vee hk \in K$

$hk \in H \implies h^{-1}hk \in H \implies k \in H \implies K \subseteq H \implies H = G$ Similarly if $hk \in K$ it follows that $K = G$, therefore atleast one of H or K must be proper.

□

Proposition 3.3. $2 < \sigma(G) \leq n - 1$

Proof. $|\Pi| \leq n - 1$ if $\Pi = \{ \langle a \rangle \mid a \text{ is non identity element of } G \}$. $\square$

Theorem 3.4 (Scorza, 1926). $\sigma(G) = 3$ if and only if there exists $N \trianglelefteq G$ such that $G/N \cong C_2 \times C_2$.

The proof follows the argument in [5], but several intermediate steps have been added for completeness and ease of comprehension, along with modification of statement aligning with [6].

Proof. ($\Leftarrow$) Suppose $N \trianglelefteq G$ and $G/N \cong C_2 \times C_2$. Let $\pi : G \rightarrow G/N$ be the natural quotient map. Let A_1, A_2, A_3 be the three proper subgroups of $C_2 \times C_2$, and define

$$H_i = \pi^{-1}(A_i), \quad i = 1, 2, 3.$$

Since the inverse image of a subgroup under a homomorphism is a subgroup, each H_i is a subgroup of G . Moreover, each A_i is proper, so $H_i \neq G$.

Now let $g \in G$. If $\pi(g) = N$, then $\pi(g) \in A_i$ for every i . Otherwise, $\pi(g)$ is one of the three nonidentity elements of $C_2 \times C_2$, each of which lies in a unique subgroup A_i . Hence $g \in H_i$ for some i . Therefore,

$$G = H_1 \cup H_2 \cup H_3,$$

so $\sigma(G) \leq 3$. Since no group is the union of one or two proper subgroups, $\sigma(G) \notin \{1, 2\}$. Hence $\sigma(G) = 3$.

($\Rightarrow$) Suppose $G = H_1 \cup H_2 \cup H_3$ with each H_i a proper subgroup, and the covering minimal.

Each H_i contains elements outside both others.

If $H_i \subseteq H_j \cup H_k$ for some permutation $\{i, j, k\}$ of $\{1, 2, 3\}$, then $G = H_j \cup H_k$, contradicting $\sigma(G) \neq 2$. So no H_i is superfluous, and in particular for each i there exist elements of H_i lying outside both remaining subgroups.

All pairwise intersections coincide.

Set $N := H_1 \cap H_2 \cap H_3$. We show $H_i \cap H_j = N$ for all $i \neq j$; it suffices to show $H_1 \cap H_2 \subseteq H_3$.

Suppose for contradiction there exists $h \in (H_1 \cap H_2) \setminus H_3$. Now pick $h_3 \in H_3 \setminus (H_1 \cup H_2)$. Consider the product $hh_3 \in G$:

- $hh_3 \notin H_1$: otherwise $h_3 = h^{-1}(hh_3) \in H_1$, since $h \in H_1$. Contradiction.
- $hh_3 \notin H_2$: otherwise $h_3 = h^{-1}(hh_3) \in H_2$, since $h \in H_2$. Contradiction.
- $hh_3 \notin H_3$: otherwise $h = (hh_3)h_3^{-1} \in H_3$, since $h_3 \in H_3$. Contradiction.

So $hh_3 \notin H_1 \cup H_2 \cup H_3 = G$, which is absurd. Hence $H_1 \cap H_2 \subseteq H_3$, giving $H_1 \cap H_2 = N$. By symmetry, $H_i \cap H_j = N$ for all $i \neq j$.

The cosets of N form a Klein four-group.

Write $X_i := H_i \setminus N$ for each i , so that as sets:

$$G = N \sqcup X_1 \sqcup X_2 \sqcup X_3.$$

We show that $\{N, X_1, X_2, X_3\}$ is closed under the operation $A \cdot B := \{ab : a \in A, b \in B\}$, and forms a group isomorphic to $C_2 \times C_2$.

(a) $X_i \cdot X_j = X_k$ for any permutation $\{i, j, k\}$ of $\{1, 2, 3\}$.

Let $x_i \in X_i$ (so $x_i \in H_i \setminus N$) and $x_j \in X_j$ (so $x_j \in H_j \setminus N$). The product $x_i x_j$ cannot lie in H_i (otherwise $x_j = x_i^{-1}(x_i x_j) \in H_i$, contradicting $x_j \notin N = H_i \cap H_j$, so $x_j \notin H_i$) and cannot lie in H_j by the symmetric argument. Hence $x_i x_j \in H_k$. Since $H_k = N \sqcup X_k$, we ask whether $x_i x_j \in N$. But $N = H_i \cap H_j$, and $x_i x_j \in H_k \setminus H_i$ (as just shown), so $x_i x_j \notin H_i \supseteq N$; thus $x_i x_j \in X_k$. This gives $X_i \cdot X_j \subseteq X_k$.

For the reverse inclusion, take any $x_k \in X_k$. Pick any $x_i \in X_i$; then $x_i x_k \in G$, and by the same argument (applied to i and k), $x_i x_k \in X_j$. So $x_k = x_i^{-1}(x_i x_k)$. Now $x_i^{-1} \in X_i$ (since $x_i \in H_i \setminus N$ implies $x_i^{-1} \in H_i \setminus N$) and $x_i x_k \in X_j$, so $x_k \in X_i \cdot X_j$. Thus $X_k \subseteq X_i \cdot X_j$, giving $X_i \cdot X_j = X_k$.

(b) $X_i \cdot X_i = N$ for each i .

From (a), $X_i \cdot X_j = X_k$ implies $X_i = X_k \cdot X_j^{-1}$; but by (a) $X_k \cdot X_j = X_i$, and since $X_j \cdot X_j^{-1} \dots$ more directly: by (a) applied twice, $X_i \cdot (X_i \cdot X_j) = X_i \cdot X_k = X_j$, and $(X_i \cdot X_i) \cdot X_j = X_j$ forces $X_i \cdot X_i = N$.

Alternatively: take $x, x' \in X_i$. Then xx'^{-1} : note $x'^{-1} \in X_i$ (as above). By (a) with $j = i$, $x \cdot x'^{-1} \in X_k$ for some k , but $xx'^{-1} \in H_i$ (since both $x, x'^{-1} \in H_i$), so $xx'^{-1} \in H_i \cap X_k$. Since $X_k = H_k \setminus N$ and $H_i \cap H_k = N$, we get $xx'^{-1} \in N$. Thus every product of two elements of X_i lands in N . Conversely, for any $n \in N$, pick $x \in X_i$; then $nx \in X_i$ (since $nx \in H_i$ and $nx \notin N$, as otherwise $x = n^{-1}(nx) \in N$), and $x \cdot (nx) = (x \cdot n \cdot x^{-1}) \cdot x^2 \dots$ instead: $n = (nx)x^{-1} \in X_i \cdot X_i$. Hence $X_i \cdot X_i = N$.

Finally for $N \leq G$ and $G/N \cong C_2 \times C_2$.

Consider the function $\phi : G/N \rightarrow C_2 \times C_2$ where we map each element of G/N as $N \mapsto (0, 0), X_1 \mapsto (0, 1), X_2 \mapsto (1, 0), X_3 \mapsto (1, 1)$ then its clearly a bijection. and we have already shown $X_i X_j = X_k$ for any permutation of $\{i, j, k\}$ of $\{1, 2, 3\}$ and $X_i X_i = N$ which respects the structure of $C_2 \times C_2$. Hence,

$$G/N \cong C_2 \times C_2. \quad \square$$

Remark 3.5. Note that in [6] V_4 is used for denoting the Klein-4 Group.

Theorem 3.6 (Haber–Rosenfeld [6, Theorem 4]). *Let p be the smallest prime divisor of $|G|$.*

- (i) *G cannot be the union of p or fewer proper subgroups.*
- (ii) *If $\Pi = \{H_i\}$ is a covering of G by $p + 1$ proper subgroups, then some H_i satisfies $[G : H_i] = p$. Moreover, if such H_i is normal in G , then every $H \in \Pi$ has index p and $p^2 \mid |G|$.*

Proof. (i) Consider the covering where $n \leq p$, $\Pi = \{H_i \mid H_i < G, 1 \leq i \leq n\}$. Since all subgroup must have identity element as common we have,

$$|G| < \sum_{i=1}^n |H_i|$$

Let H be the subgroup with highest order, and since p is the least prime divisor we have

$$[G : H] \leq p \implies \frac{|G|}{|H|} \leq p \implies \frac{|G|}{p} \leq |H|$$

Then we have, $|G| < n|H| \implies |G| < n \cdot \frac{|G|}{p} \implies p < n$ which is a contradiction.

(ii) Consider the covering with $p + 1$ subgroups $\Pi = \{H_i \mid H_i < G, 1 \leq i \leq p + 1\}$

Suppose none of the subgroups have index p . then $[G : H_i] \geq p + 1 \implies |H_i| \geq \frac{|G|}{p+1}$

Take the subgroup H_i with maximum order then $|G| < (p + 1)|H_i| \implies |G| < (p + 1) \frac{|G|}{p+1} \implies |G| < |G|$ is a contradiction. So, one of the subgroup must have index p .

Let H be the subgroup with index p and assume its normal.

Since H is normal then HH_i is a subgroup of G with $H, H_i \subseteq HH_i$. If HH_i is proper, we will find the covering with p subgroups by excluding H, H_i and including HH_i which contradicts (i). So $HH_i = G$.

$$|G| = |HH_i| = \frac{|H||H_i|}{|H \cap H_i|} \implies |H_i| = |G| \frac{|H \cap H_i|}{|H|} \implies |H_i| = p|H \cap H_i|$$

Now let the indices of other H_i be $q_i > p$, and for convenience let $H_1 = H$.

$$\begin{aligned} |G| &\leq |H_1| + \sum_{i=2}^{p+1} |H_i| - |H_i \cap H_1| = |H_1| + \sum_{i=2}^{p+1} \frac{|G|}{q_i} - \frac{|G|}{pq_i} \\ &< |H_1| + p \left(\frac{|G|}{p} - \frac{|G|}{p^2} \right) = \frac{|G|}{p} + |G| - \frac{|G|}{p} = |G| \implies |G| < |G| \end{aligned}$$

Which is a contradiction. So $p = q_i$. Finally, $[G : H_i \cap H_1] = [G : H_i][H_i : H_i \cap H_1] = p \cdot \frac{|G|}{|H_1|} = p \cdot \frac{|G|}{p} = p^2 \implies p^2$ divides $|G|$.

□

3.1.1 The covering sum inequality

Lemma 3.7. *If $G = \bigcup_{i=1}^n H_i$ is a covering by proper subgroups, then*

$$\sum_{i=1}^n \frac{1}{[G : H_i]} \geq 1.$$

Proof. For each $g \in G$, since $g \in H_i$ for some i , summing the indicator functions $\mathbf{1}_{H_i}$ over G and switching the order of summation gives $\sum_{i=1}^n |H_i| \geq |G|$. Dividing by $|G|$ yields the result. $\square$

Lemma 3.8. *If $N \trianglelefteq G$, then $\sigma(G) \leq \sigma(G/N)$.*

Proof. If G/N is cyclic the inequality is trivial, so assume $\sigma(G/N) = n < \infty$. Let $G/N = \bar{H}_1 \cup \cdots \cup \bar{H}_n$ be a minimal covering by proper subgroups. Define $H_r := \pi^{-1}(\bar{H}_r)$ for each r , where $\pi : G \rightarrow G/N$ is the natural projection.

By the Correspondence Theorem, each H_r is a subgroup of G with $[G : H_r] = [G/N : \bar{H}_r] \geq 2$, so each H_r is proper. For any $g \in G$, $\pi(g) \in \bar{H}_r$ for some r , hence $g \in H_r$. Thus $G = H_1 \cup \cdots \cup H_n$, giving $\sigma(G) \leq n = \sigma(G/N)$. $\square$

Lemma 3.9. *For any finite groups H and K , $\sigma(H \times K) \leq \min\{\sigma(H), \sigma(K)\}$. If additionally $\gcd(|H|, |K|) = 1$, then equality holds.*

Proof. Inequality. If $\sigma(H) = n < \infty$, say $H = \bigcup_{i=1}^n A_i$ with each $A_i \subsetneq H$, then $H \times K = \bigcup_{i=1}^n (A_i \times K)$ and each $A_i \times K$ is a proper subgroup of $H \times K$. So $\sigma(H \times K) \leq \sigma(H)$, and symmetrically $\sigma(H \times K) \leq \sigma(K)$, giving the inequality.

Equality when $\gcd(|H|, |K|) = 1$. It remains to show $\sigma(H \times K) \geq \min\{\sigma(H), \sigma(K)\}$. We use the following structural fact:

We claim :If $\gcd(|H|, |K|) = 1$, every subgroup $L \leq H \times K$ satisfies $L = \pi_H(L) \times \pi_K(L)$, where π_H, π_K are the coordinate projections.

Proof of Claim. Let $(h, k) \in L$. Since $\gcd(|H|, |K|) = 1$, the orders $|h|$ and $|k|$ are coprime, so by Bezout's theorem there exist integers s, t with $s|h| + t|k| = 1$. Then:

$$(h, k)^{t|k|} = (h^{t|k|}, e) = (h^{1-s|h|}, e) = (h, e),$$

so $(h, e) \in L$, and symmetrically $(e, k) \in L$. Hence $L \supseteq \pi_H(L) \times \pi_K(L)$; the reverse inclusion is immediate from the definition of projection. $\square$

Now let $H \times K = \bigcup_{i=1}^n L_i$ be any covering by proper subgroups, with $n = \sigma(H \times K)$. By the Claim, write $L_i = A_i \times B_i$ where $A_i = \pi_H(L_i) \leq H$ and $B_i = \pi_K(L_i) \leq K$. Projecting the covering onto H :

$$H = \pi_H \left(\bigcup_{i=1}^n L_i \right) = \bigcup_{i=1}^n A_i.$$

Since each $L_i \leq H \times K$, either $A_i \leq H$ or $B_i \leq K$. Let $I = \{i : A_i \leq H\}$. The subgroups $\{A_i\}_{i \notin I}$ all equal H and are redundant in the union $\bigcup_i A_i = H$, so $\{A_i\}_{i \in I}$ already covers H . Hence $\sigma(H) \leq |I| \leq n$.

By a symmetric argument with $J = \{i : B_i \leq K\}$, we get $\sigma(K) \leq n$. Since $I \cup J = \{1, \dots, n\}$ (every L_i is proper), this gives:

$$\min\{\sigma(H), \sigma(K)\} \leq n = \sigma(H \times K).$$

Combined with the inequality direction, $\sigma(H \times K) = \min\{\sigma(H), \sigma(K)\}$. □

These lemmas were discussed in [3].

3.2 Equal covering theorems

Since the proofs of the following theorems can be immediately found in [6] we did not bother to mention it, but we formalized them. See 4

Theorem 3.10 ([6, Theorem 13]). *If $\Pi = \{H_i\}$ is an equal covering of G , then $\exp(G) \mid |H_i|$ for all i .*

Corollary 3.11 ([6, Corollary 1]). *If $\exp(G) \nmid |K|$ for every maximal subgroup K of G , then G has no equal covering.*

Theorem 3.12 ([6, Theorem 14]). *D_{2n} has an equal covering if and only if n is even. When n is even, $\Pi = \{\langle r \rangle, \langle r^2, s \rangle, \langle r^2, rs \rangle\}$ is an equal covering.*

Theorem 3.13 ([6, Theorem 15]). *Every non-cyclic finite p -group has an equal covering.*

Theorem 3.14 ([6, Theorems 16, 17]). (i) *If G or H has an equal covering, so does $G \times H$.*

(ii) *Every non-cyclic finite nilpotent group has an equal covering.*

Theorem 3.15 ([6, Theorem 18]). *If $|G| = p_1 p_2 \cdots p_k$ with p_i distinct primes, then G has no equal covering.*

Theorem 3.16 ([6, Theorem 19, Corollary 4]). (i) *If $N \trianglelefteq G$ and G/N has an equal covering, so does G .*

(ii) *If $G = H \rtimes K$ and K has an equal covering, so does G .*

4 Lean 4 Formalization

Lean is an open-source programming language and proof assistant that enables correct, maintainable, and formally verified code. Lean is interactive theorem prover based on dependent type theory designed especially for mathematics and software verification, with its core type theory expressive

[2]

4.1 Overview

We formalized the following results in Lean 4:

- **Theorem 1.** A group G has a covering if and only if G is non-cyclic (both directions, including an explicit construction of the covering $\{\langle x \rangle : x \in G\}$ in the reverse direction).
- **Theorem 2.** No group is the union of two proper subgroups, i.e. $\sigma(G) \neq 2$.
- **Theorem 13.** If $\Pi = \{H_i\}$ is an equal covering of G , then $\exp(G) \mid |H_i|$ for every $H_i \in \Pi$.
- **Corollary 1.** If $\exp(G) \nmid |K|$ for every maximal subgroup K of G , then G has no equal covering.
- **Theorem 16.** If G has an equal covering, then so does $G \times H$ for any group H (via an explicit product-subgroup construction).
- **Theorem 19.** If $f : G \twoheadrightarrow Q$ is a surjective homomorphism and Q has an equal covering, then G has an equal covering, obtained by pulling the covering back along f .

Theorems 15 and 17 (every non-cyclic p -group, resp. nilpotent group, has an equal covering) are currently stated as `axioms` with the correct type signature, pending a full proof; see §??.

4.2 Key Mathlib tools

Rather than importing Mathlib's `Subgroup` API directly, we built a small self-contained algebraic core mirroring it, so that the intended correspondence to Mathlib is explicit:

- `Group`, `IsSubgroup`, and `Subgroup` play the role of Mathlib's `Group` and `Subgroup`, with membership given by a predicate `subset :  $\alpha \rightarrow \text{Prop}$` rather than a `SetLike` coercion.

- `Covering` and `EqualCovering` package a family of proper subgroups indexed by `i : I`, together with the covering property (`covers`) and, for equal coverings, a proof that all members have the same `Size`.
- `order_subgroup`, `exp`, and `order_elem` stand in for Mathlib’s `Nat.card` and `Monoid.exponent`, together with an axiomatized `lagrange` lemma (order of an element divides the order of any subgroup containing it) and `exp_divides` (the exponent is the least common multiple of element orders, expressed via divisibility).
- `MaxSubgroup` and `divides_max` formalize maximality and the fact that every subgroup embeds in some maximal subgroup, used for Corollary 1.
- `DirectProduct` and `ProdSubgroup` construct the product group $G \times H$ and lift a subgroup of G to a subgroup of $G \times H$, used in Theorem 16.
- `Homomorphism`, `Surjective`, and `PreimageSubgroup` formalize group homomorphisms and preimages of subgroups under a surjection, used in Theorem 19.

[Describe: `Subgroup`, `Subgroup.index`, `Subgroup.comap`, `Finset`, relevant API lemmas.]

4.3 Challenges

- **Axiomatizing rather than importing.** To keep the formalization self-contained and readable, we defined our own `Group/Subgroup` structures instead of using Mathlib’s, which meant re-axiomatizing standard facts (Lagrange’s theorem, the exponent–order relationship) rather than reusing proofs already in Mathlib. This trades a shorter file for a formalization that is not yet literally checked against Mathlib’s group theory.
- **Order as a type, not a number.** We initially represented the order of a subgroup with an abstract `Size : Subgroup G → Type u` to sidestep finiteness assumptions; this was later replaced by a numeric `order_subgroup : Subgroup G → Nat` for the finite-group results (Theorem 13 onward), which is closer to Mathlib’s `Nat.card` but required threading finiteness hypotheses through every subsequent lemma.
- **Case-heavy classical proofs.** Results like Scorza’s theorem (Theorem 3.4, $\sigma(G) = 3 \iff G/N \cong C_2 \times C_2$) rely on case splits over coset membership that are easy to write on paper but require careful use of `Classical.byContradiction` and explicit witness extraction in Lean, since the underlying group is not assumed decidable.

- **Quotients and the Correspondence Theorem.** Theorem 19 required formalizing preimages of subgroups under a homomorphism from scratch (`PreimageSubgroup`), since we do not yet have quotient groups formalized in this file; a full treatment of Theorem 19(ii) (via $G = H \rtimes K$) is left for future work.

4.4 Representative Lean code

The proof that no group is the union of two proper subgroups (Theorem 2) illustrates the style of the formalization:

Listing 1: Formalization of Theorem 2: $\sigma(G) \neq 2$

```
theorem theorem2 {alpha : Type u} (G : Group alpha) (H K : Subgroup G)
  (pH : IsProper H) (pK : IsProper K) :
  Not (forall x, H.subset x or K.subset x) := by
  intro covers
  rcases pH with <k, hk_not_H>
  rcases pK with <h, hh_not_K>
  have hk_K : K.subset k := by
    cases covers k with
    | inl h_in => contradiction
    | inr k_in => exact k_in
  have hh_H : H.subset h := by
    cases covers h with
    | inl h_in => exact h_in
    | inr k_in => contradiction
  have h_mul_k := covers (G.mul h k)
  cases h_mul_k with
  | inl h_in =>
    have k_in_H := cancel_lemma H h k hh_H h_in
    contradiction
  | inr k_in =>
    have h_in_K := cancel_lemma2 K h k hk_K k_in
    contradiction
```

The proof that equal coverings force the exponent of G to divide the order of every member (Theorem 13) shows how `Lagrange` and `exp_divides` compose:

Listing 2: Formalization of Theorem 13

```
theorem theorem13 {alpha : Type u} {G : Group alpha} {I : Type u}
  (eqCov : FiniteEqualCovering G I) (i : I) :
  exp G Divides order_subgroup (eqCov.cov.sub i) := by
```

```

apply exp_divides
intro x
have <j, x_in_j> := eqCov.cov.covers x
have lag := lagrange (eqCov.cov.sub j) x x_in_j
have eq_sz := eqCov.equalSz j i
rw [← eq_sz]
exact lag

```

Listing 3: Formalization of the covering sum inequality

Find remaining theorems on github.

4.5 Scope and limitations

[Honest statement of what is and is not formalized, and why.]

5 GAP Computational Investigation

GAP is a system for computational discrete algebra, with particular emphasis on Computational Group Theory. [1]

5.1 The algorithm

We modified the code provided by [6] to test and find the equal coverings of finite Groups.

Listing 4: GAP algorithm for equal covering detection

```

1 Algorithm EqualCovering(G)
2 S := AllSubgroups(G);
3 D := DivisorsInt(Order(G));
4
5 orders := [];
6 for d in D do
7     if d < Order(G) and RemInt(d, Exponent(G)) = 0 then
8         Append(orders, [d]);
9     fi;
10 od;
11
12 truthvalues := [];
13
14 for d in orders do
15     union := [];

```

```

16
17     for s in S do
18         if Order(s) = d then
19             Append(union, ShallowCopy(Elements(s)));
20         fi;
21     od;
22
23     Append(truthvalues, [Elements(G) = Set(union)]);
24 od;
25
26 return true in truthvalues;

```

5.2 Results

Group ID	Name	Exp	Nilp.?	Equal cov.?	Reason
[61, 1]	C_{61}	61	Yes	No	T1 (cyclic)

5.3 Challenges

- Its not feasible to extend the checking for finite simple groups. Velasquez [6] already checked it up to order 10000. Checking on the larger group would be very slow.
- They program was slow when order of group increased, we had to optimize.

5.4 Patterns observed

Overall the results were consistent with the theorems. The simple groups we checked were also not equally coverable, adding further evidence to the conjecture.

6 The Simple Group Conjecture

6.1 Statement and known evidence

Conjecture 6.1 ([6, p. 21]). *If G is a finite simple group, then G has no equal covering.*

The paper establishes this computationally for all finite simple groups of order less than 100,000 (Table 3 of [6]).

6.2 Theoretical analysis

Proposition 6.2 ([6, Proposition 2]). *If G is a finite non-cyclic simple group with $\exp(G) = |G|/2$, then G has no equal covering.*

Remark 6.3. Since simple groups have no non-trivial normal subgroups, Theorem 3.16 cannot be used to transfer equal coverings from quotients. Any equal covering of a simple group must arise entirely from its internal subgroup structure, subject to the exponent constraint of Theorem 3.10.

6.2.1 The alternating groups A_n

The *alternating group* on n letters, denoted by A_n , is the subgroup of the symmetric group S_n consisting of all even permutations:

$$A_n = \{\sigma \in S_n : \text{sgn}(\sigma) = 1\}.$$

Since the sign homomorphism

$$\text{sgn} : S_n \rightarrow \{\pm 1\}$$

is surjective with kernel A_n , it follows that

$$|A_n| = \frac{n!}{2}.$$

For $n \geq 5$, the alternating group A_n is non-abelian and simple; that is, its only normal subgroups are the trivial subgroup and A_n itself. Consequently, the groups A_n ($n \geq 5$) form an infinite family of finite non-abelian simple groups.

We attempt to investigate the conjecture 1.1 for such groups.

.....

Proposition 6.4. *For any prime $p \geq 5$, the alternating group A_p has no equal covering.*

Proof.

□

7 Conclusion

We have extended the study of equal coverings of finite groups in three directions. Theoretically, we provided complete proofs of the Haber–Rosenfeld theorem and all core theorems of [6], and obtained a partial theoretical result (Theorem 6.4) toward Theorem 1.1. We formalized [list] of these results in Lean 4. Computationally, we extended the GAP classification to order 120, producing a complete table consistent with all theoretical predictions.

Future directions.

- Complete the Lean formalization of Theorems.
- Target Alternating groups of prime letters with the conjecture (≥ 5 .)

References

- [1] Groups, algorithms and programming : <https://www.gap-system.org/>.
- [2] Lean 4 <https://lean-lang.org/>.
- [3] J.H.E Cohn. On n -sum groups.
- [4] Seymour Haber and Azriel Rosenfeld. Groups as unions of proper subgroups. *Source: The American Mathematical Monthly*, 66:491–494.
- [5] Gaetano Scorza. Bollettino unione matematica italiana.
- [6] Andrew Velasquez-Berroteran. Equal coverings of finite groups. 6 2022.